

Ali Osman Berk Şapcı

Curriculum Vitae

Franklin Antonio Hall, University of California, San Diego

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Research Interests

Computational biology; metagenomics, phylogenetics and phylogenomics, and sequence analysis. Scalable algorithms for large-scale genome analysis, probabilistic data structures, and probabilistic modeling for molecular sequences.

Education

Ph.D. in Bioinformatics and Systems Biology

University of California, San Diego

Advisor: [Siavash Mirarab](#)

2022–present

La Jolla, CA, USA

M.Sc. in Computer Science and Engineering

Sabancı University

Thesis: *Automated Behavioral Analysis of Drosophila melanogaster Sleep*

Advisors: [Öznur Taştan](#), [Sündüz Keleş](#) (co-advisor)

2020–2022

Istanbul, Turkey

B.Sc. in Computer Science and Engineering, Minor in Mathematics

Sabancı University

Graduation Project: *Active Learning for Turkish Named Entity and Event Recognition*

Advisors: Reyhan Yeniterzi, [Öznur Taştan](#)

2015–2020

Istanbul, Turkey

Research Experience

University of California, San Diego

Graduate Student Researcher — Advisor: [Siavash Mirarab](#)

2022–present

La Jolla, CA, USA

La Jolla Institute of Immunology

Graduate Student Researcher — Supervisor: [Ferhat Ay](#)

Fall 2022

La Jolla, CA, USA

Sabancı University

Graduate Research Assistant — Advisors: [Öznur Taştan](#), [Sündüz Keleş](#)

2020–2022

Istanbul, Turkey

University of Wisconsin–Madison

Intern Research Assistant — Supervisor: [Sündüz Keleş](#)

Summer 2020

[remote]

Technical University of Munich

Intern Research Assistant — Supervisor: [Martin Schreiber](#)

Summer 2019

Munich, Germany

Sabancı University

Undergraduate Research Assistant — Supervisor: [Kamer Kaya](#)

2018–2019

Istanbul, Turkey

Teaching Experience

University of California, San Diego

Teaching Assistant

ECE 208: Computational Evolutionary Biology

ECE 30: Introduction to Computer Engineering

La Jolla, CA, USA

Spring 2026

Spring 2025

Sabancı University

Teaching Assistant

CS 300: Data Structures

CS 301: Algorithms

CS 404: Artificial Intelligence

CS 302: Formal Languages and Automata Theory

Istanbul, Turkey

Spring 2022

Fall 2021

Spring 2021

Fall 2020

Sabancı University

Undergraduate Teaching Assistant

CS 301: Algorithms

CS 201: Introduction to Computing

Istanbul, Turkey

Spring 2020

Fall 2018, Spring 2019

Publications

Preprints & Manuscripts in Preparation

1. Şapcı, A. O. B., & Mirarab, S. (2026). Gdiff: genome-wide distance-based pattern matching via differentiation.
2. Weng, Y., McDonald, D., Zhao, J., Iter, M., Alabiad, A., Patel, L., Allaband, C., Shepherd, I., Hansen, Şapcı, A. O. B., A., ... & Knight, R. (2026). Web of Life 3: a fine-resolution bacterial and archaeal genome resource for scalable long-read metagenomics.
3. Arasti, S., Şapcı, A. O. B., Rachtman, E., El-Kebir, M., & Mirarab, S. (2026). Deconvolving phylogenetic distance mixtures. *bioRxiv*. doi:10.64898/2026.01.18.700179

Journal Publications

4. Şapcı, A. O. B., Arasti, S., Braun, E. L., & Mirarab, S. (2026). Phlag: Scalable detection of genomics regions with unexplained phylogenetic heterogeneity. *Bioinformatics*, 42 (Supplement 1), btag273. doi:10.1093/bioinformatics/btag273 (ISMB Proceedings)
5. Melendez, D., Şapcı, A. O. B., Bafna, V., & Mirarab, S. (2026). SPruCE: Utilizing ultraconserved elements of DNA for population-level genetic diversity estimation. *Molecular Ecology Resources*, 26(3), e70145. doi:10.1111/1755-0998.70145
6. Şapcı, A. O. B., & Mirarab, S. (2026). krepp: A k -mer-based maximum pseudo-likelihood method for estimating read distances and genome-wide phylogenetic placement. *Genome Biology*, 27(1), 108. doi:10.1186/s13059-026-03999-y
7. Keleş, M. F., Şapcı, A. O. B., Brody, C., Palmer, I., Mehta, A., Ahmadi, S., Le, C., Taştan, Ö., Keleş, S., & Wu, M. N. (2025). FlyVISTA, an integrated machine learning platform for deep phenotyping of sleep in *Drosophila*. *Science Advances*, 11(11), eadq8131. doi:10.1126/sciadv.adq8131
8. Şapcı, A. O. B., & Mirarab, S. (2024). Memory-bound k -mer selection for large & evolutionary diverse reference libraries. *Genome Research*, 34(9), gr.279339.124. doi:10.1101/gr.279339.124 (RECOMB Proceedings)
9. Şapcı, A. O. B., Rachtman, E., & Mirarab, S. (2024). CONSULT-II: Accurate taxonomic identification and profiling using locality-sensitive hashing. *Bioinformatics*, 40(4), btae150. doi:10.1093/bioinformatics/btae150
10. Şapcı, A. O. B., Lu, S., Yan, S., Ay, F., Taştan, Ö., & Keleş, S. (2023). MuDCoD: Multi-subject community detection in personalized dynamic gene networks from single-cell RNA sequencing. *Bioinformatics*, 39(10), btad592. doi:10.1093/bioinformatics/btad592
11. Şapcı, A. O. B., Kemik, H., Yeniterzi, R., & Taştan, Ö. (2023). Focusing on potential named entities during active label acquisition. *Natural Language Engineering*, 29(6), 1–23. doi:10.1017/S1351324923000165

Peer-Reviewed Conference Proceedings

12. Şapcı, A. O. B., & Mirarab, S. (2025). Alignment-free estimation of read-to-genome distances and its applications. In *Research in Computational Molecular Biology (RECOMB)*, pp. 316–320. Springer Nature Switzerland. doi:10.1007/978-3-031-90252-9_30
13. Şapcı, A. O. B., & Mirarab, S. (2024). Memory-bound and taxonomy-aware k -mer selection for ultra-large reference libraries. In *Research in Computational Molecular Biology (RECOMB)*, pp. 340–343. Springer Nature Switzerland. doi:10.1007/978-1-0716-3989-4_26
14. Şapcı, A. O. B., Rachtman, E., & Mirarab, S. (2023). CONSULT-II: Taxonomic identification using locality sensitive hashing. In *Comparative Genomics (RECOMB-CG)*, pp. 196–214. Springer Nature Switzerland. doi:10.1007/978-3-031-36911-7_13 (Best Paper Award)
15. Şapcı, A. O. B., Taştan, Ö., & Yeniterzi, R. (2020). Active learning for Turkish text classification. In *28th Signal Processing and Communications Applications Conference (SIU)*, pp. 1–4. doi:10.1109/SIU49456.2020.9302289

Fellowships and Awards

Awards

ISMB Conference Fellowship	2026
RECOMB Travel Fellowship	2026
IMSI Workshop Travel Support	2025
RECOMB-CG Best Paper Award	2023
Highest Ranking Student Award, Turkish Council of Higher Education	2015

Fellowships

Powell Fellowship, The Charles Lee Powell Foundation	2022
Fulbright Ph.D. Grant, Fulbright Commission (<i>declined</i>)	2022
Full Graduate Scholarship, Sabancı University	2020–2021
Sakıp Sabancı Fellowship, Sabancı University	2015–2020
DEVAK Scholarship, Deloitte Education Foundation	2018–2020

Selected Presentations

<i>Proceedings Talk, ISMB, Washington, D.C.</i>	July 2026
Phlag: Scalable detection of genomics regions with unexplained phylogenetic heterogeneity.	
<i>Proceedings Talk, RECOMB, Thessaloniki, Greece</i>	May 2026
Deconvolving phylogenetic distance mixtures.	
<i>Short Talk, RECOMB-seq, Thessaloniki, Greece</i>	May 2026
A genome-wide likelihood framework for distance-based pattern matching.	
<i>Invited Lecture, BIO 310, Sabancı University</i>	May 2026
Scalable methods for phylogeny-aware sequence analysis.	
<i>Tutorial, IMSI, Chicago</i>	August 2025
Read-to-genome distance calculation and phylogenetic placement using krepp.	
<i>Proceedings Talk, RECOMB, Seoul, South Korea</i>	April 2025
Estimating read-to-genome distances from homologous k -mers.	
<i>Proceedings Talk, RECOMB, Boston, MA</i>	April 2024
Memory-bound and taxonomy-aware k -mer selection.	
<i>Short Talk, BioSys, in conjunction with ASPLOS, La Jolla, CA</i>	April 2024
Memory-bound and taxonomy-aware k -mer selection.	
<i>Proceedings Talk, RECOMB-CG, Istanbul, Türkiye</i>	April 2023
CONSULT-II: Taxonomic identification using locality sensitive hashing.	
<i>Spotlight Talk, MLCB</i>	November 2021
MuDCoD: Multi-subject community detection in dynamic gene networks.	

Posters

<i>Research in Computational Molecular Biology (RECOMB), Thessaloniki, Greece</i>	May 2026
A genome-wide likelihood framework for distance estimation and pattern matching.	
<i>Probabilistic Modeling in Genomics (ProbGen), UC Berkeley</i>	March 2026
A probabilistic framework for sequence distances & genome-wide (phylogenetic) pattern matching.	
<i>Institute for Mathematical and Statistical Innovation, Chicago</i>	August 2025
Mapping strongly discordant regions on the genome using hidden Markov models.	
<i>Machine Learning in Computational Biology (MLCB)</i>	November 2021
MuDCoD: Multi-subject community detection in dynamic gene networks.	

Professional Activities

Organizer

ISMB Tutorial: “Metagenomic sequence analysis using k -mer based methods”

July 2026

Reviewer *(includes sub- and co-reviewing)*

- *Journals:* Molecular Ecology Resources; Genome Research; Proceedings of Machine Learning Research; Bioinformatics; Bioinformatics Advances; Machine Learning.
- *Conferences:* RECOMB (2025, 2026); ISMB (2022); ACML (2021, 2022).

Mentor

Undergraduates and M.Sc. students

- Isaac Ang (2026–present)
- Ali Alabiad (2024–present)
- Sankalp Kumaraswamy (2024–2025)

References

Siavash Mirarab, Professor, Electrical & Computer Engineering
University of California, San Diego

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Vineet Bafna, Professor, Computer Science & Engineering
University of California, San Diego

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Edward L. Braun, Professor, Biology
University of Florida

ebraun68@ufl.edu

Sunduz Keles, Professor, Statistics and Biostatistics
University of Wisconsin–Madison

keles@stat.wisc.edu